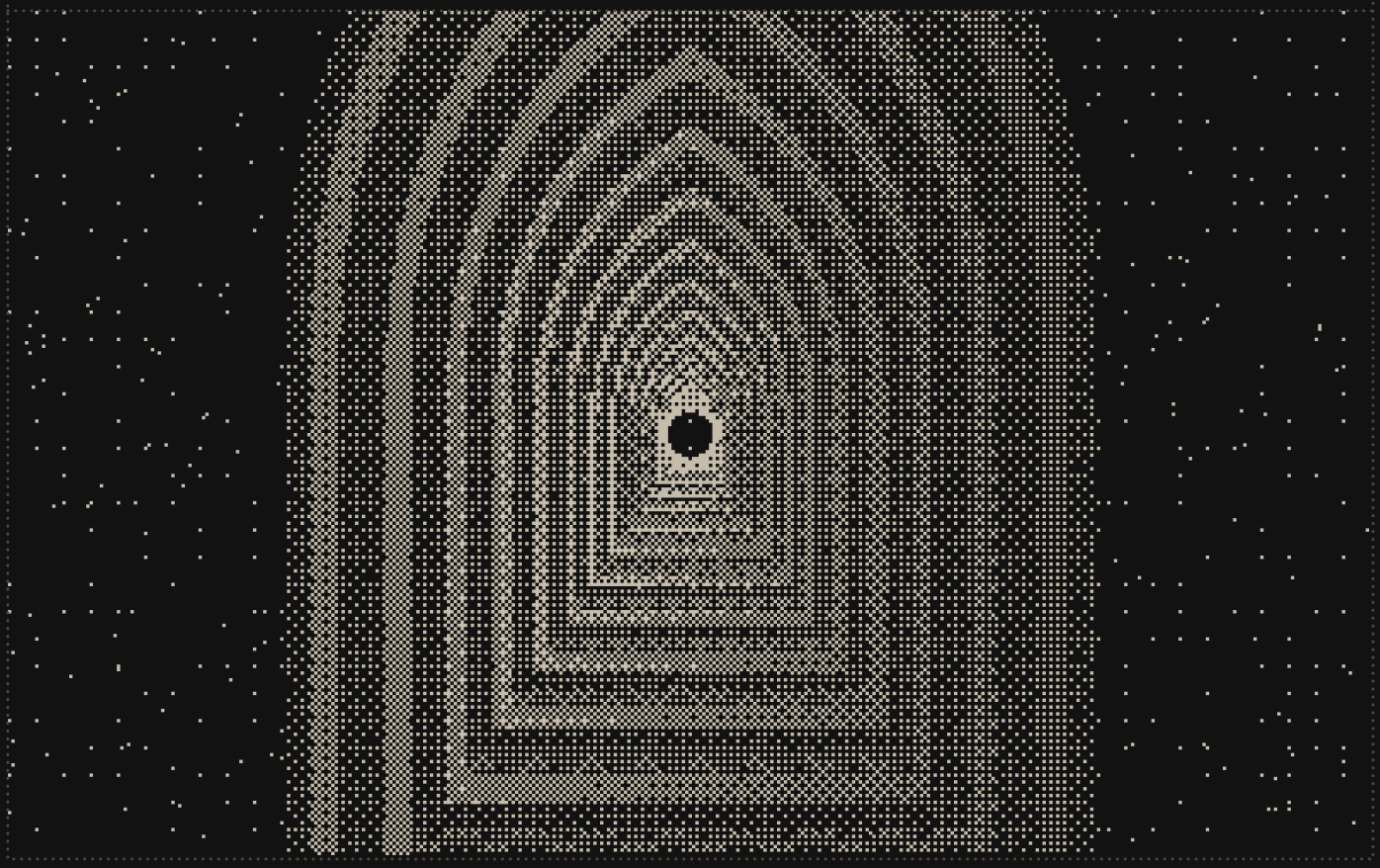
PLATE // ZENO'S NAVE



KAELEARN // ANALYSIS I // FOLIO 03

Cauchy Sequences

How to prove a sequence arrives without knowing where it is going.

FOLIO	03
SUBJECT	Analysis I
SOURCES	Specimen: no real sources
READ	2 min view · 14 min folio
FILED	2026-10-11



PART I

The trouble with limits

To prove that a sequence converges, the definition you were given makes a strange demand: name the limit first. In

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N : \; |a_n - L| < \varepsilon$$

the L sits right in the middle, and you are expected to bring it. For $a_n = 1/n$ that costs nothing. For $a_n = \sum_{k=1}^{n} 1/k^2$ you would need to know the answer is $\pi^2/6$, which took Euler years. Most sequences do not arrive with their limit printed on the box.

Cauchy's move: stop asking *where* the terms are going. Ask only whether they are getting close to *each other*. That you can check with nothing but the terms themselves.

Cauchy wrote it down in 1821. Bolzano had it four years earlier, in Prague, and nobody read him. I read him.

THE TWO-MINUTE VIEW // READ THIS FIRST

- **The problem.** The ε–N definition needs the limit L in advance.
- **The fix.** A Cauchy sequence *(Cauchy-Folge)*: past some point, all terms huddle, $\forall \varepsilon > 0 \; \exists N \; \forall m, n \geq N : |a_m - a_n| < \varepsilon$.
- **Convergent ⇒ Cauchy**, always, by one triangle inequality.
- **Cauchy ⇒ convergent** holds in $\mathbb{R}$ and fails in $\mathbb{Q}$. This *is* completeness *(Vollständigkeit)*.
- **The proof** runs Cauchy ⇒ bounded ⇒ a convergent subsequence (Bolzano–Weierstrass) ⇒ the whole sequence follows it.
- **The trap.** $|a_{n+1} - a_n| \to 0$ is not enough: the harmonic sums take ever smaller steps and still walk to infinity.

GUESS FIRST // WRONG IS FINE, GUESSING IS THE POINT

01 If consecutive terms get arbitrarily close, $|a_{n+1} - a_n| \to 0$, must the sequence converge?

02 The decimal truncations of $\sqrt{2}$, namely $1,\ 1.4,\ 1.41,\ 1.414, \ldots$, all live in $\mathbb{Q}$. Do they converge *in* $\mathbb{Q}$?

03 Can you prove $\sum 1/k^2$ converges without knowing its value? How would you even start?

Terms · *Begriffe*

ENGLISH	DEUTSCH	IN ONE LINE
Cauchy sequence	*Cauchy-Folge*	all terms past N within ε of each other
complete	*vollständig*	every Cauchy sequence converges
subsequence	*Teilfolge*	(a_{n_k}) with $n_1 < n_2 < \cdots$
bounded	*beschränkt*	$\lvert a_n \rvert \le M$ for all n
accumulation point	*Häufungspunkt*	the limit of some subsequence
triangle inequality	*Dreiecksungleichung*	$\lvert x+y \rvert \le \lvert x \rvert + \lvert y \rvert$

PART II

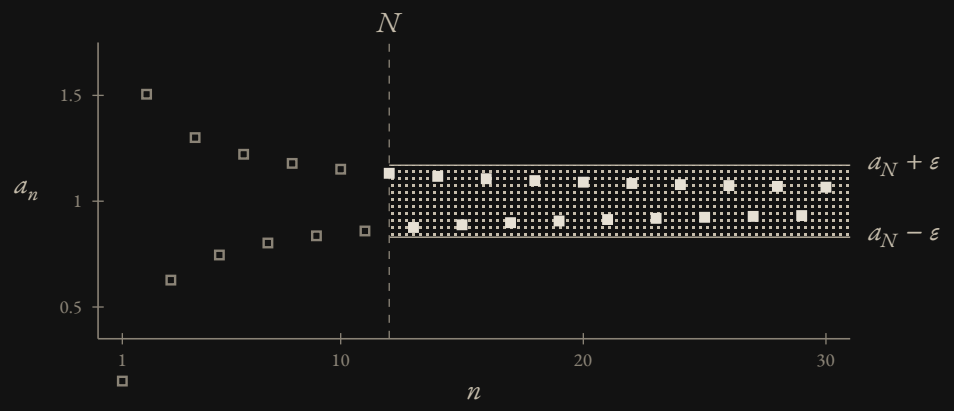
The definition

DEFINITION 03.1 // CAUCHY SEQUENCE · CAUCHY-FOLGE

A sequence (a_n) of real numbers is a *Cauchy sequence* if

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall m, n \ge N : \quad \lvert a_m - a_n \rvert < \varepsilon.$$

Read the quantifiers in order, because the order is the content. First someone hands you an ε, as small as they like. Then you choose N, and N may depend on ε. Then *every* pair of terms past N must sit within ε of each other. Not neighbours. Every pair: the thousandth and the 10^{100}th.



Past N every term stays within ε of a_N, so any two are within 2ε of each other. The band is centred on a term, not on the limit. You never needed L.

THEOREM 03.2 // THE CAUCHY CRITERION · CAUCHY-KRITERIUM

A sequence of real numbers converges if and only if it is a Cauchy sequence.

PROOF IN ONE BREATH $\Rightarrow$: two terms near L are near each other, $|a_m - a_n| \le |a_m - L| + |L - a_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$. $\Leftarrow$: Cauchy $\Rightarrow$ bounded $\Rightarrow$ Bolzano–Weierstrass hands you a subsequence with a limit $L \Rightarrow$ the huddle drags every term to L.

Why does the $\Leftarrow$ direction need Bolzano–Weierstrass?

Why?

Being Cauchy says nothing about *where* the terms are, only that they huddle. Something has to produce a candidate L, a point to huddle around. Bolzano–Weierstrass is that something, and it is exactly where the completeness of $\mathbb{R}$ enters. In $\mathbb{Q}$ the huddle can form around a hole: the decimals of $\sqrt{2}$ huddle, and $\mathbb{Q}$ has nothing at the centre.

Why is $|a_{n+1} - a_n| \to 0$ not enough?

Why?

It controls neighbours, not distant pairs. The harmonic sums $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ move by $\frac{1}{n+1} \to 0$ per step, yet $H_{2n} - H_n \ge \frac{1}{2}$ forever. Tiny steps, infinite walk.

If you keep one sentence: Cauchy trades a destination for a crowd.

TRIGGERS // IF YOU SEE THIS, REACH FOR THAT

YOU SEE	REACH FOR
Convergence to prove, no candidate limit	the Cauchy criterion
$\lvert a_m - a_n \rvert$ with both indices free	let $m > n$, telescope: $\sum_{k=n}^{m-1} \lvert a_{k+1} - a_k \rvert$
$\lvert a_{n+1} - a_n \rvert \le q \lvert a_n - a_{n-1} \rvert$, $q < 1$	a geometric tail, $\le \frac{q^n}{1-q} \lvert a_1 - a_0 \rvert$
A series $\sum b_k$	Cauchy for partial sums: $\lvert \sum_{k=n+1}^{m} b_k \rvert < \varepsilon$



FOR THE DECK // MAKE THESE YOURSELF, THE MAKING IS THE LEARNING

DEF	Cauchy sequence: the definition, quantifiers in order.	$\forall \varepsilon > 0 \; \exists N \; \forall m, n \ge N : \; \lvert a_m - a_n \rvert < \varepsilon$
CLOZE	In {{$\mathbb{R}$}}, a sequence converges iff it is {{Cauchy}}.	The first gap is the hypothesis that fails in $\mathbb{Q}$.
WHY	Why does Cauchy $\Rightarrow$ convergent fail in $\mathbb{Q}$?	The decimals of $\sqrt{2}$ are Cauchy in $\mathbb{Q}$; the point they huddle round is missing.
TRIGGER	Convergence to prove, limit unknown. Reach for?	The Cauchy criterion, usually via a telescoping or contraction bound.